Unit 2 FRQ Set 3: Coaching Guide

Use this in two passes. **Pass 1:** for any part you couldn't finish, read only the 'Coach' questions and try again (a different colour pen). **Pass 2:** grade both attempts against the rubric.

Question	Max	Attempt 1 (timed)	After coaching
Q1 MR	10		
Q2 TBR	12		
Q3 LAB	10		
Q4 QQT	8		
Total	**40**		

Rubrics follow the College Board's style ('1 point for…'). They're written for practice. They aren't official scoring guidelines, but they test the same skills.

Question 1: Mathematical Routines

10 points

A small ball of mass m on a string of length L is swung in a vertical circle. At the bottom of the circle the ball has speed v_b; at the top it has speed v_t.

Part A (2 pts)

COACH: ASK YOURSELF

- At the top, which way does the string pull? (Toward the center, always.)
- Is there a 'centripetal force' arrow? (No.)

A FULL-CREDIT ANSWER



Top: tension DOWN (toward the center) and gravity down. Bottom: tension UP and gravity down, with T longer than mg.

Top: T and mg both downward	1 pt
Bottom: T up, mg down, T longer	1 pt

WHERE POINTS LEAK Tension up at the top (a string can only pull toward the pivot), or adding a centripetal arrow.

Part B (3 pts)

COACH: ASK YOURSELF

- Which way is the center from the bottom? Make that +.
- Which force points toward the center, and which points away?

A FULL-CREDIT ANSWER

Toward the center (up) +: $\Sigma F = mv^2/r$ → $T - mg = mv_b^2/L$ → **$T = mg + mv_b^2/L$**

Newton's 2nd law with centripetal acceleration	1 pt
Correct signs (T toward the center, mg away)	1 pt
Correct expression	1 pt

WHERE POINTS LEAK $T = mv^2/L$, which forgets that gravity is working against the string at the bottom.

Part C (2 pts)

COACH: ASK YOURSELF

- (ii) 'Minimum speed at the top' happens when which force becomes zero?

A FULL-CREDIT ANSWER

(i) $T = (0.20)(9.8) + (0.20)(4.0)^2/0.80 = 1.96 + 4.0 =$ **6.0 N**.

(ii) Top, with $T = 0$: $mg = mv^2/L \rightarrow v = \sqrt{(gL)} = \sqrt{(9.8 \times 0.80)} =$ **2.8 m/s**.

Correct tension at the bottom with units	1 pt
Correct minimum speed $\sqrt{(gL)} \approx 2.8$ m/s	1 pt

WHERE POINTS LEAK Setting $v = 0$ at the top for the minimum. Then the ball just falls, so it isn't moving in a circle.

Part D (3 pts)

COACH: ASK YOURSELF

- Compare T at the top ($mv^2/L - mg$) with T at the bottom ($mv^2/L + mg$). Where is the ball also moving fastest?

A FULL-CREDIT ANSWER

At the bottom. There, the string has to supply the centripetal force AND cancel gravity ($T = mg + mv^2/L$), while at the top gravity helps ($T = mv^2/L - mg$). The ball is also moving fastest at the bottom. Max speed: $8.0 = 1.96 + 0.20v^2/0.80 \rightarrow v^2 = 24.2 \rightarrow$ **$v \approx 4.9$ m/s**.

Bottom	1 pt
Justifies with gravity opposing the string at the bottom (or with the expressions)	1 pt
Correct maximum speed ≈ 4.9 m/s	1 pt

WHERE POINTS LEAK 'At the top, because gravity pulls it down', which has it the wrong way round.

THINK FURTHER Try it (carefully, outside!) with a bucket of water. Why doesn't the water fall out at the top? (When $v \geq \sqrt{gL}$, gravity alone isn't enough to provide the needed centripetal force, so the bucket has to push on the water.)

Question 2: Translation Between Representations

12 points

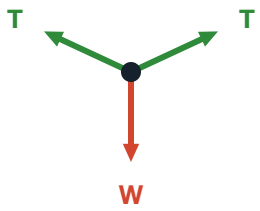
A sign of weight W hangs at rest from a light ring. Two identical cables hold the ring, each making angle θ with the horizontal, as shown.

Part A (3 pts)

COACH: ASK YOURSELF

- Three things pull on the ring. Which ones?
- Cables pull ALONG themselves, away from the ring.

A FULL-CREDIT ANSWER



Two equal tensions along the cables (up and out at angle θ), and the sign's pull W straight down.

Two tensions along the cables, away from the ring	1 pt
W downward	1 pt
Tensions drawn equal; arrows at the correct angles	1 pt

WHERE POINTS LEAK Drawing the tensions horizontal or vertical instead of along the cables.

Part B (3 pts)

COACH: ASK YOURSELF

- Which components hold the sign up? sin or cos? (θ is measured from the HORIZONTAL.)
- Equilibrium in y.

A FULL-CREDIT ANSWER

Equilibrium: $\Sigma F_y = 0 \rightarrow 2T \sin\theta - W = 0 \rightarrow$ **$T = W / (2 \sin\theta)$**. ($\Sigma F_x = 0$ is automatic by symmetry.)

Uses equilibrium (ΣF = 0) in the vertical direction	1 pt
Correct vertical components T sin θ (both cables)	1 pt
Correct expression	1 pt

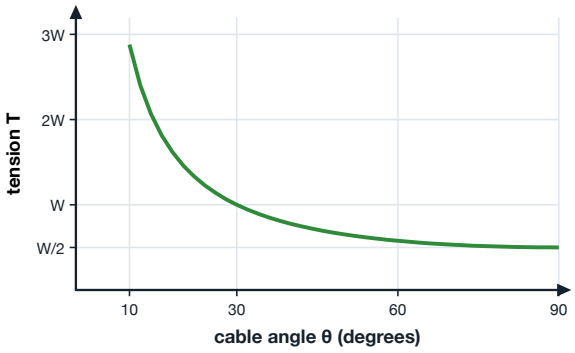
WHERE POINTS LEAK Using cos θ. Draw the triangle: the side opposite θ (measured from the horizontal) is the vertical one.

Part C (4 pts)

COACH: ASK YOURSELF

- What is T when θ = 90° (cables straight up)?
- What happens as θ gets small? Does T stay finite?

A FULL-CREDIT ANSWER



Decreasing curve	1 pt
Concave up, rising steeply at small θ	1 pt
T = W/2 at 90°, labelled	1 pt
Horizontal (or flattening) slope near 90°, and no zero or negative values	1 pt

WHERE POINTS LEAK Drawing a straight line. 1/sin θ isn't linear.

Part D (2 pts)

COACH: ASK YOURSELF

- What is sin 0°? What does that do to T = W/(2 sin θ)?

A FULL-CREDIT ANSWER

As θ → 0, sin θ → 0, so T = W/(2 sin θ) → ∞ (the graph shoots up). With horizontal cables there's no vertical component left to hold up the weight, so no finite tension can keep the sign in equilibrium. The cables always sag a little.

Uses the expression or graph: T grows without limit as θ → 0	1 pt
Explains that horizontal cables have no vertical component to balance W	1 pt

WHERE POINTS LEAK Just saying 'it's impossible'. You need to point to the math or the graph.

THINK FURTHER Unequal angles: left cable at 30°, right cable at 60°. Which cable has the bigger tension? (The steeper one, at 60°. It carries more of the load.)

Question 3: Experimental Design and Analysis

10 points

Students want to determine the gravitational field strength g using a low-friction cart on an adjustable ramp. They have a meterstick, a protractor, a stopwatch and a motion sensor.

Part A (2 pts)

COACH: ASK YOURSELF

- Frictionless ramp: a = ? (from your FBD).
- How could you get a with only a meterstick and a stopwatch?

A FULL-CREDIT ANSWER

1. Set the ramp to an angle θ (measured with the protractor) and release the cart from rest.
2. Measure its acceleration: use the slope of the motion sensor's v–t graph, or time how long it takes to roll a measured distance d from rest and use $a = 2d/t^2$.
3. Repeat for at least 5 angles, with 3 trials each, and average.

Varies θ and measures it	1 pt
A valid method for measuring a, with repeated trials	1 pt

WHERE POINTS LEAK Not saying the cart starts from REST when using $d = \frac{1}{2}at^2$.

Part B (2 pts)

COACH: ASK YOURSELF

- Derive a for a frictionless ramp. Is a proportional to θ, or to sin θ?

A FULL-CREDIT ANSWER

Frictionless ramp: $a = g \sin\theta$. Fill in the sin θ column (0.087, 0.174, 0.259, 0.342, 0.423). Graph **a vs. sin θ**, and the slope is g.

Fills in sin θ (or uses it)	1 pt
Graphs a vs. sin θ, slope = g	1 pt

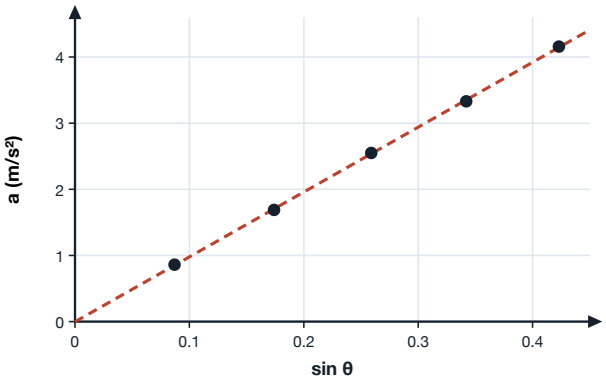
WHERE POINTS LEAK Graphing a vs. θ. It looks almost linear at small angles, but the slope isn't g.

Part C (4 pts)

COACH: ASK YOURSELF

- sin θ has no units. That's fine. Just label it 'sin θ'.

A FULL-CREDIT ANSWER



Axes labelled (with units for a)	1 pt
Appropriate scale	1 pt
Points plotted correctly	1 pt
Best-fit line	1 pt

WHERE POINTS LEAK Using the angle in degrees on the x-axis after saying you'd graph sin θ. Be consistent with part B.

Part D (2 pts)

COACH: ASK YOURSELF

- Slope from two points on the line.
- What force did we ignore that would make a smaller than g sin θ?

A FULL-CREDIT ANSWER

Slope ≈ **9.8 m/s²**, so g ≈ 9.8 m/s². A small amount of friction in the wheels or air resistance makes each a slightly smaller than g sin θ, which lowers the measured slope.

g from the slope of the best-fit line, with units	1 pt
Valid physical reason for a low value (friction or drag)	1 pt

WHERE POINTS LEAK 'Human error' is too vague. Name the physics: friction acts opposite the motion and reduces the acceleration.

THINK FURTHER If the motion sensor were tilted so it read every acceleration 0.10 m/s² too high, what changes, the slope or the intercept? (The intercept. The slope, and so g, is unaffected.)

Question 4: Qualitative/Quantitative Translation

8 points

Skater A (40 kg) and Skater B (80 kg) stand at rest on frictionless ice, 6.0 m apart, holding the ends of a light rope. Skater B pulls the rope hand over hand until the skaters meet.

Part A (3 pts)

COACH: ASK YOURSELF

- B is doing the pulling. Does that make the force on A bigger? (3rd law!)
- Take both skaters as the system. Are there any external horizontal forces? So what happens to the center of mass?

A FULL-CREDIT ANSWER

The rope pulls on A and on B with forces of equal size (3rd law). Equal forces give the lighter skater, A, the larger acceleration, so A covers more distance before they meet. Another way to see it: with no external horizontal force, the system's center of mass stays still, and it sits closer to the heavier skater. They meet at the center of mass, **closer to B's starting position**.

Equal forces on the two skaters	1 pt
Lighter A accelerates more / moves farther (or the center of mass stays fixed)	1 pt
Concludes: closer to B's start	1 pt

WHERE POINTS LEAK 'Closer to A, because B is pulling', which assumes the puller moves less. Who pulls doesn't matter.

Part B (3 pts)

COACH: ASK YOURSELF

- Put A at x = 0 and B at x = 6.0 m. Where is the center of mass?
- Alternative: the ratio of accelerations is the ratio of distances (same time, both from rest).

A FULL-CREDIT ANSWER

$x_{cm} = (40 \cdot 0 + 80 \cdot 6.0)/(40 + 80) = 480/120 =$ **4.0 m from A** (2.0 m from B).

Check: $a_A/a_B = 80/40 = 2$, so A travels twice as far: 4.0 m + 2.0 m = 6.0 m ✓

Center-of-mass equation (or the acceleration ratio from 3rd + 2nd laws)	1 pt
Correct substitution	1 pt
4.0 m from A	1 pt

WHERE POINTS LEAK Getting 2.0 m: you measured from the wrong skater. Always say 'from A'.

Part C (2 pts)

COACH: ASK YOURSELF

- Is 4.0 m from A closer to A or to B?
- Does the center-of-mass calculation care who pulls?

A FULL-CREDIT ANSWER

4.0 m from A is 2.0 m from B, so the skaters meet closer to B's start, as argued in part A. **No change** if A pulls: the forces are still a 3rd-law pair, the external force is still zero, and the center of mass still stays at 4.0 m.

Connects 4.0 m to 'closer to B'	1 pt
No change if A pulls, with a reason (3rd law or fixed center of mass)	1 pt

WHERE POINTS LEAK Treating the two parts as unrelated. A QQT always wants the words and the math to point at each other.

THINK FURTHER Repeat with Skater B on rough ice (friction on B only). Does the meeting point move? (Yes: now there's an external force, so the center of mass is no longer fixed.)

Error log: FRQ Set 3

For each miss or guess: what went wrong, using one of these types: **C** concept (didn't know the physics) · **R** representation (FBD/graph wrong) · **M** math/algebra · **F** FRQ wording (right idea, missing the words) · **T** time.

Q	Type	What I did vs. what I should have done	Re-try date / ✓

Score: _____ / 40 Sure-and-right: _____ Time used: _____